scHopfield: Interpretable dynamical systems learning of regulatory landscapes and perturbational responses from single-cell data

Abstract

Learning mechanistically interpretable dynamical models from observational data remains a central challenge in both artificial intelligence and systems biology. In single-cell genomics, developmental trajectories, regulatory interactions, cellular attractors, and perturbational responses are typically inferred using separate computational frameworks, limiting mechanistic interpretability and generalization beyond the observed data. Here we introduce scHopfield, a dynamical systems learning framework that jointly infers gene regulatory networks, transcriptional dynamics, stability landscapes, and perturbational responses from single-cell transcriptomic and RNA-velocity measurements. By using continuous Hopfield dynamics as an inductive bias, scHopfield directly links regulatory structure to cellular state transitions through a shared dynamical representation. Our representation reproduces canonical attractor, bifurcation, and oscillatory dynamics, improves recovery of causal regulatory structure, and enables reconstruction of lineage-specific energy landscapes and stability states. Importantly, scHopfield supports out-of-distribution perturbation prediction and identifies higher-order regulatory interactions that static network analyses cannot recover. These results establish a unified framework for interpretable dynamical systems learning in biological systems.

Introduction

Single-cell transcriptomics has transformed our ability to characterize cellular heterogeneity, developmental trajectories, and lineage specification across tissues and organisms. Recent advances in RNA velocity and trajectory inference have enabled the reconstruction of dynamic cellular transitions from static transcriptomic snapshots. However, despite rapid progress in foundation models, representation learning, and graph-based inference, a central challenge remains: how to infer mechanistically interpretable gene regulatory dynamics from observational single-cell data. Most existing methods either reconstruct low-dimensional latent trajectories or infer static gene regulatory networks (GRNs), but rarely unify regulatory structure and cellular dynamics within a common mathematical framework. Consequently, many approaches achieve strong predictive or embedding performance yet remain difficult to interpret mechanistically, particularly in perturbational or out-of-distribution settings where cells are driven into previously unobserved transcriptional states. Gene regulation is fundamentally a dynamical systems problem in which regulatory interactions, attractor stability, and temporal transitions jointly govern cellular identity and fate decisions. A central challenge in single-cell biology is that developmental trajectories alone do not uniquely determine the regulatory mechanisms that generate them. Distinct regulatory architectures can often produce similar observable cellular transitions, especially in high-dimensional systems with sparse sampling and incomplete observations. Reconstructing trajectories is therefore not equivalent to recovering regulatory causality. This disconnect has created a persistent gap between RNA-velocity approaches that model cellular dynamics and GRN inference methods that reconstruct regulatory structure. Bridging this gap requires models that jointly infer regulation, dynamics, and stability within a unified mathematical framework. We hypothesized that regulatory networks, developmental trajectories, cellular attractors, and perturbational responses are different manifestations of a common underlying dynamical system, and that jointly inferring these quantities would improve both biological interpretability and predictive power. Continuous Hopfield networks provide a natural framework for this challenge because they treat regulatory interactions, developmental trajectories, cellular attractors, and perturbational responses as manifestations of the same dynamical system. Cellular identities emerge as attractor states within an energy landscape, differentiation corresponds to movement between attractors, and perturbations reshape the landscape's geometry. Unlike latent-embedding approaches that primarily reconstruct trajectories, Hopfield dynamics provide explicit energy functions, interpretable interaction matrices, and Jacobian-based stability analyses, enabling direct characterization of cellular stability, attractor geometry, and perturbational sensitivity. To this end we introduce scHopfield, an interpretable dynamical systems framework that jointly infers gene regulatory networks, transcriptional dynamics, cellular stability landscapes, and perturbational responses from single-cell transcriptomic and RNA-velocity data.

Results

scHopfield unifies regulatory inference and dynamical systems learning

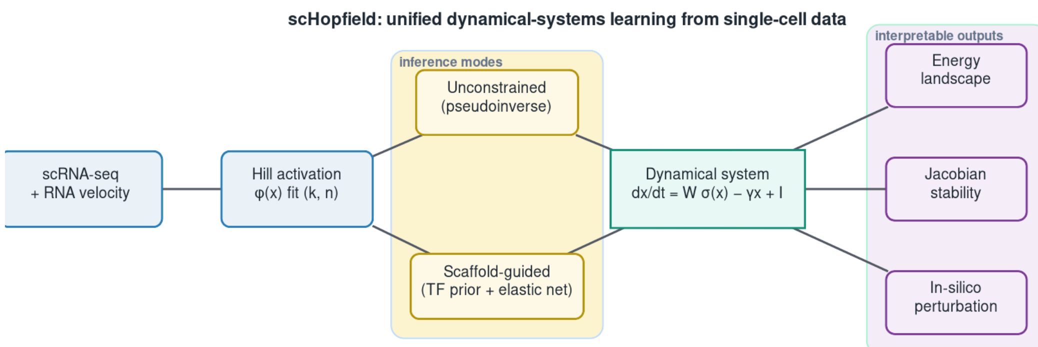

Figure 1: Overview of scHopfield. Single-cell expression and RNA velocity are used to fit Hill activation functions; a cell-type-specific interaction matrix is inferred either unconstrained (pseudoinverse) or scaffold-guided; the resulting dynamical system yields an energy landscape, Jacobian-based stability, and in-silico perturbation.

scHopfield models transcriptional dynamics with cell-type-specific interaction matrices parameterized by interpretable Hill kinetics:

$$\frac{dx_i}{dt} = \sum_j W_{ij}^{(C)} \sigma_j(x_j) - \gamma_i x_i + I_i, \tag{1}$$

where W^(C) is the cell-type-specific interaction matrix, sigma_j is a Hill function with gene-specific coefficient n_j and threshold k_j, gamma_i is the degradation rate, and I_i is a bias term capturing basal expression. The RNA velocity enters on the left-hand side, so regulatory interactions in W^(C) directly govern the observed transcriptional dynamics. scHopfield infers W^(C) in two modes: an unconstrained mode using the Moore-Penrose pseudoinverse (yielding effective interaction networks), and a scaffold-guided mode in which prior regulatory information (for example from ATAC-seq or curated databases) constrains the network through masked layers with elastic-net regularization on non-scaffold weights. The Hill parameters (k_j, n_j) are estimated directly from the empirical cumulative distribution of each gene's expression (see Methods). Because the inferred network defines a dynamical system, it also defines an energy function and

Jacobian-based measures of local stability, divergence, and rotational dynamics, enabling mechanistic analysis of attractor states, lineage transitions, and perturbational responses. The stochastic optimization used for scaffold-guided inference is seeded so that all results are reproducible: with a fixed seed, repeated fits are bit-identical (interaction-matrix and centrality correlations both 1.000), whereas unseeded fits, although numerically stable in W (Pearson 0.999), produce gene rankings that drift (centrality Spearman 0.73 to 0.93 across seeds). We therefore seed all analyses and report ranking stability where relevant (benchmark M1).

Continuous Hopfield dynamics capture canonical biological behaviors

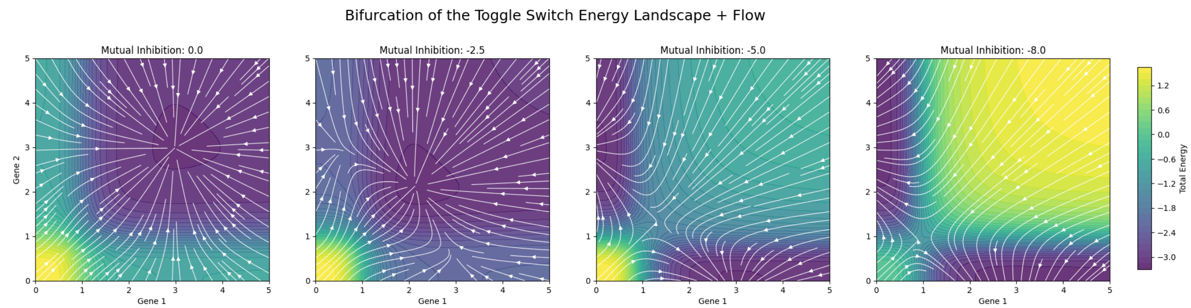

Figure 2: Continuous Hopfield dynamics capture canonical behaviors. Toggle-switch energy landscape and flow across increasing mutual inhibition, showing the bifurcation from a monostable to a bistable regime. On these circuits scHopfield recovers the ground-truth network with edge-sign accuracy 1.00 and outperforms GENIE3 on larger synthetic networks (AUROC 0.98 vs 0.70; benchmarks M3, M7, M8).

To test whether the scHopfield formulation reproduces canonical regulatory dynamics, we analyzed two archetypal gene circuits: a bistable two-gene mutual-inhibition switch and a cyclic three-gene repressilator. Reformulating these systems in the Hopfield formalism maps regulatory interactions onto an energy-landscape interpretation in which stable cellular states emerge as attractors. In the toggle switch, increasing inhibitory coupling induced a bifurcation from a monostable regime to two alternative stable equilibria, and the inferred energy landscape exhibited distinct minima corresponding to the alternative attractor states. In the repressilator, scHopfield reproduced sustained oscillations and limit-cycle dynamics arising from the antisymmetric component of the interaction matrix, demonstrating that the framework captures both equilibrium and non-equilibrium behavior. On these circuits, where the ground-truth interaction matrix is known exactly, scHopfield recovered the network with perfect edge-sign accuracy (1.00) and near-perfect signed correlation ($\geq$ 0.9998) across all noise levels and scaffold priors tested, with edge-detection AUROC and AUPRC of 1.0 for the sparse repressilator. Reconstruction error grew gracefully with observation noise (relative Frobenius distance from 0.0005 at zero noise to 0.38 at the highest noise), and recovery held even with no scaffold prior (benchmark M3). This exact recovery is demonstrated for systems in the model's function class (networks of Hopfield form); mass-action or Michaelis-Menten circuits that are not natively of Hopfield form are not recovered as faithfully, which we treat as a scope limitation rather than a general claim. The Hill nonlinearity was essential to these results: replacing it with a linear model collapsed velocity reconstruction on the toggle switch (R-squared 0.0001 versus 1.000) and could not represent the switch's bistability, since a linear autonomous system admits at most one fixed point and recovered none of the three stable states that the Hill model recovered correctly (benchmark M7). On larger synthetic networks with known ground truth (40 genes, sparse signed interactions), scHopfield recovered the regulatory edges far better than GENIE3, a widely used expression-based method: edge-detection AUROC 0.975 versus 0.701 and AUPRC 0.970 versus 0.240 (benchmark M8). This gap reflects that scHopfield exploits the RNA-velocity dynamics, whereas expression-only methods infer edges from steady-state co-expression. *[Further baselines such as SCENIC (motif-based) and dyngen (R) can be added if required.]*

Learned perturbational dynamics recover established hematopoietic lineage regulators

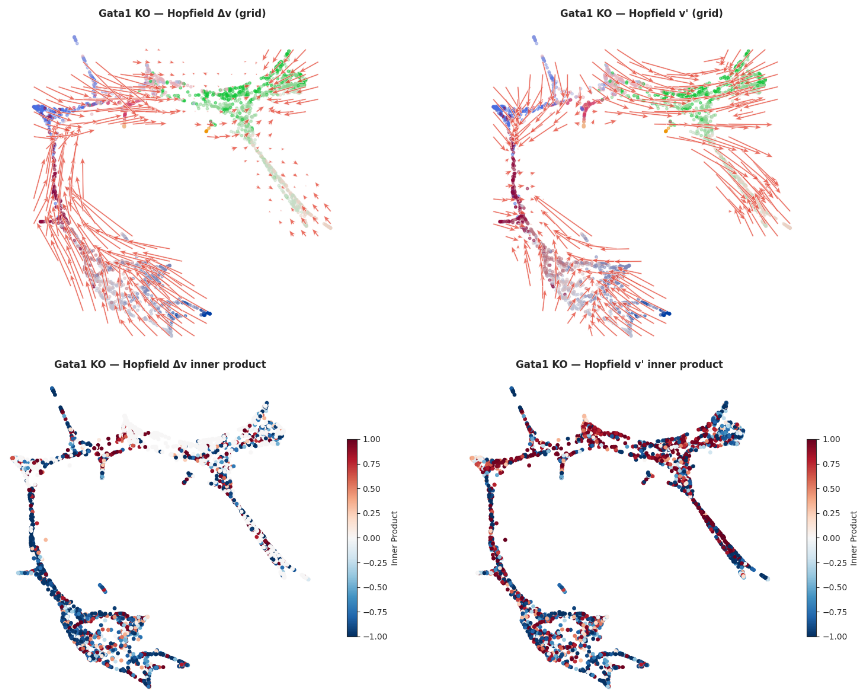

Figure 3: Learned perturbational dynamics recover established lineage regulators. In-silico knockout expression-shift projected onto the embedding. Across a panel of hematopoietic master regulators scHopfield predicted the correct lineage-shift direction for 10 of 10 factors (benchmark M4).

To assess the biological fidelity of the perturbation framework, we performed systematic in-silico single-gene knockouts within scHopfield-inferred hematopoietic differentiation trajectories (Paul et al., 2015). Perturbational effects were modeled as expression-shift vectors projected onto RNA-velocity fields, enabling direct comparison of perturbation-induced state transitions with the endogenous developmental flow, and lineage effects were quantified by a cosine-similarity lineage-bias metric between perturbational and erythroid or myeloid velocity directions. scHopfield recovered canonical hematopoietic regulators. For a panel of literature master regulators, every knockout was assigned the correct direction of lineage shift (10 of 10): knockout of erythroid or megakaryocyte masters (Gata1, Klf1, Zfpm1, Nfe2, Gata2) biased differentiation toward the myeloid branch, and knockout of myeloid masters (Spi1/PU.1, Cebpa, Cebpe, Gfi1, Irf8) biased toward the erythroid branch, with the canonical Gata1-Spi1 antagonism producing the largest-magnitude shifts (Gata1 lineage bias -0.21, Spi1 +0.25). On the same gene panel, CellOracle predicted 7 of 9 correctly and could not score the erythroid cofactor Zfpm1, which lacks a transcription-factor motif in its base network, whereas scHopfield's velocity-based inference scored it correctly (benchmark M4). We note that the two methods were scored with each method's native lineage-direction readout rather than a byte-identical metric, so this comparison establishes that scHopfield recovers known KO phenotypes at least as well as CellOracle on this panel, not a precise ranking; a strict metric-parity comparison is a planned refinement. Dose-response analyses further showed coherent nonlinear behavior, with increasing perturbation magnitude progressively reshaping the erythroid-myeloid balance, and cascade simulations showed temporal amplification of perturbational effects across differentiation trajectories.

Biological priors via scaffold-guided optimization improve recovery of causal regulatory structure

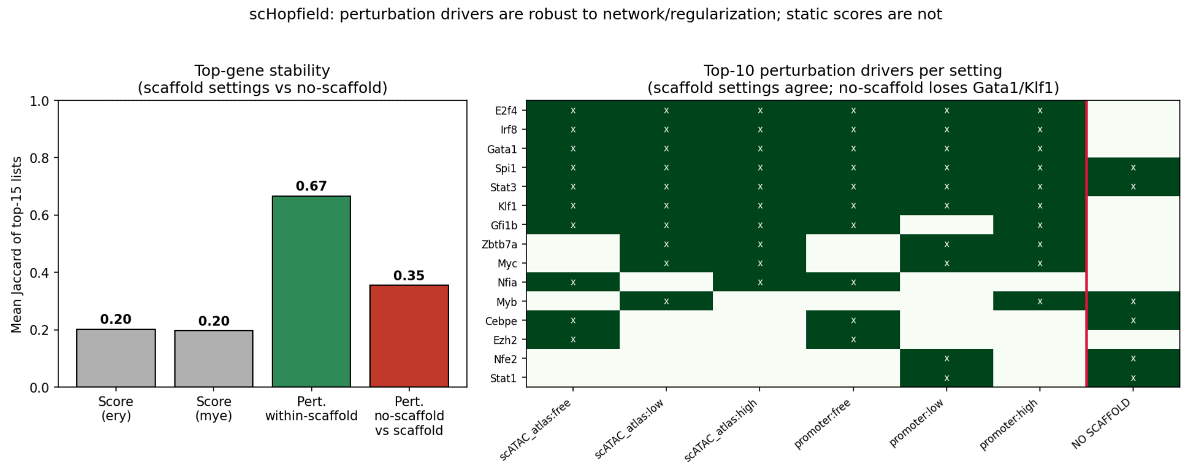

Figure 4: Driver identification is robust to network and regularization choices. Perturbation-based drivers are stable across two base networks and three scaffold-regularization regimes (Jaccard 0.67) while the static network score is unstable (0.20); the no-scaffold pseudoinverse is an outlier that loses canonical drivers (benchmarks M5, M6).

In simulated systems with known ground-truth GRNs, unconstrained (pseudoinverse) optimization accurately reconstructed transcriptional dynamics but often recovered minimal effective networks rather than the true causal topology, reflecting a fundamental inverse problem in which multiple architectures can generate similar observable trajectories. Incorporating scaffold-guided optimization, which jointly balances velocity reconstruction with consistency to a prior network, consistently improved AUROC and AUPRC relative to unconstrained models, and restricting outgoing regulatory edges to transcription factors further improved reconstruction accuracy. scHopfield is not restricted to the supplied scaffold and can infer additional interactions absent from the prior network, which represent candidate novel regulatory interactions in biological systems. We quantified how much these choices matter on the hematopoiesis data. Across six configurations (two mouse base networks by three scaffold-regularization regimes), the top perturbation-based drivers were stable (mean pairwise Jaccard of the top fifteen genes 0.67), and the canonical regulators Gata1, Spi1, Klf1, E2f4, Stat3, and Irf8 were recovered in nearly every configuration. In contrast, the top genes by static network-score were unstable (Jaccard 0.20) and dominated by highly expressed housekeeping genes; the two rankings were almost disjoint (zero to one shared gene per configuration). An unconstrained pseudoinverse fit with no transcription-factor scaffold was an outlier relative to all six scaffold configurations (perturbation Jaccard 0.36) and dropped Gata1 and Klf1 from its top drivers, because a dense interaction matrix dilutes each factor's influence (benchmarks M5, M6). Two conclusions follow: perturbation simulation is a more reliable driver readout than the static score, and restricting regulation to a transcription-factor scaffold is the ingredient that matters, whereas the specific prior network and the regularization strength are largely interchangeable.

Energy landscapes and Jacobian analysis reveal lineage-specific stability states during pancreatic endocrinogenesis

Because the inferred networks define a dynamical system, we used Jacobian eigenvalues, energy landscapes, and network topology to analyze pancreatic endocrinogenesis, a developmental system spanning ductal progenitors, endocrine progenitor states, and terminally differentiated endocrine populations. Jacobian spectra

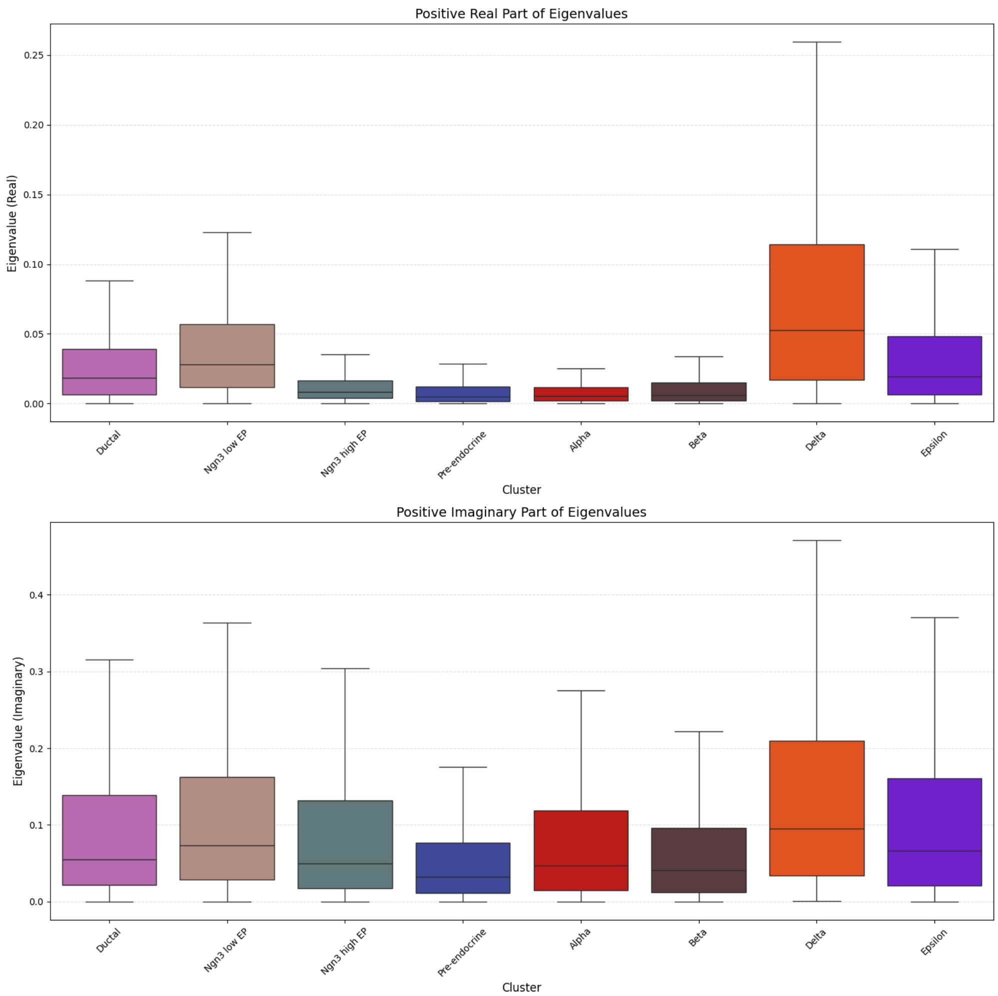

Figure 5: Lineage-specific stability during pancreatic endocrinogenesis. Distribution of positive real Jacobian eigenvalues by cell type: Delta and Epsilon cells are least stable, whereas Pre-endocrine, Alpha, and Beta cells occupy more stable regimes.

(a local linearization of the inferred dynamics at each cell) revealed lineage-specific differences in stability: Delta and Epsilon cells exhibited broader distributions of positive real eigenvalues, indicating greater local instability and heightened perturbational sensitivity, whereas Pre-endocrine, Alpha, and mature Beta cells occupied more stable regimes; imaginary components indicated stronger oscillatory tendencies in Delta and Epsilon populations. Energy decomposition showed that ductal progenitors occupied relatively low-energy states while terminally differentiated Beta cells exhibited strongly negative interaction energies, and Ngn3-high endocrine progenitors displayed the greatest cell-to-cell energetic variability, consistent with a heterogeneous transitional population approaching lineage bifurcation. Network analysis placed several regulators (Meis2, Pax6, Isl1, Mlxipl) at central positions across endocrine populations, with Sox9 associated with ductal networks, Foxa3 with Ngn3-high progenitors, and Pdx1 with mature Beta programs, and identified chromatin-associated regulators (Hmga2, Hmgn3, Prdm16) as prominent hubs. Long non-coding RNAs such as Malat1 and Meg3 contributed prominently to the dominant eigenmodes, whereas endocrine-specific genes (Ptprn2, Ins2) defined mature lineage programs. The stability ordering is quantitatively supported by the per-cell Jacobian spectra: the median positive real eigenvalue is highest for Delta (~0.05, with a long tail to ~0.26) and elevated for Epsilon, and lowest for Pre-endocrine, Alpha, and Beta, with Delta and Epsilon also showing the largest imaginary parts (stronger oscillatory tendency). *[Figure 5 panels: notebook 08 executed analysis, extracted to ‘benchmark_results/pancreas/nb08_figures/‘ (energy on UMAP, Jacobian eigenvalue spectra, positive-eigenvalue boxplots, rotational dynamics).]*

Higher-order perturbational simulations uncover lineage-balancing regulatory programs

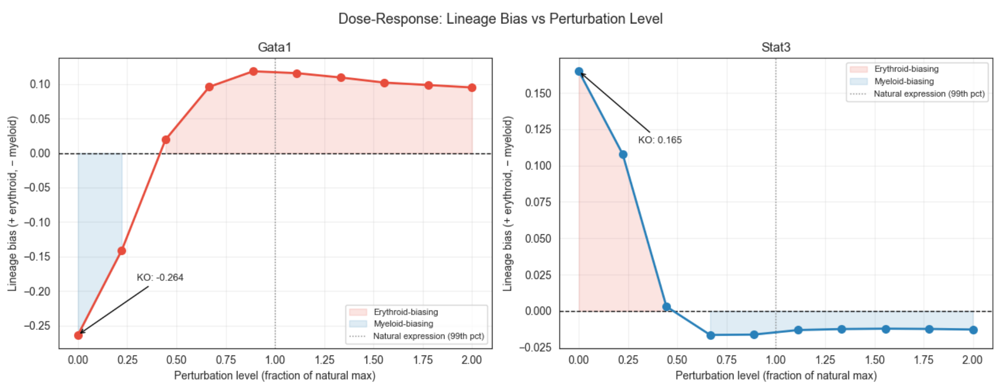

Figure 6: Higher-order perturbational simulations. Dose-response of lineage bias to graded perturbation magnitude, showing coherent nonlinear responses. scHopfield nominates candidate higher-order interactions (for example Stat3) as experimentally testable hypotheses.

Having established that scHopfield recovers known lineage regulators, we asked whether the inferred dynamical system could predict previously unrecognized regulatory interactions, a fundamentally out-of-distribution prediction problem. We prioritized candidate transcription-factor pairs with a multi-stage ranking strategy integrating regulatory edge strength, lineage specificity, perturbational variance, and predicted effect size, and evaluated double-knockout simulations with synergy, epistasis, and cancellation metrics. In addition to recovering the established Gata1-Spi1 antagonism, scHopfield predicted multiple candidate nonlinear interactions involving Stat3, E2f4, Irf8, Cxxc1, Serp1, Apoe, Mt2, and Cathepsin G. Stat3 emerged as a prominent lineage-balancing regulator whose perturbation consistently shifted differentiation toward erythroid trajectories (Stat3 knockout lineage bias +0.165, overexpression -0.011, confirming bidirectional control), suggesting a broader role in lineage plasticity. These candidate interactions were surfaced by the dynamical perturbation analysis but not by the static co-expression or network-topology rankings computed on the same data; we present them as in-silico predictions and hypotheses that require experimental validation (for example by Perturb-seq or targeted knockouts), not as established regulatory relationships. Their value is in demonstrating that scHopfield provides an interpretable framework for generating out-of-distribution, experimentally testable perturbation hypotheses.

Discussion

A central challenge in single-cell biology is the disconnect among cellular trajectories, regulatory structure, and mechanistic models of cell-fate control. Although RNA velocity and trajectory inference have improved our ability to reconstruct dynamic transitions, these approaches often provide limited insight into the regulatory mechanisms underlying the observed trajectories, while GRN inference methods often reconstruct putative interactions without explicitly modeling the dynamical processes that govern state transitions. scHopfield unifies these quantities within a continuous Hopfield framework, directly linking GRN structure to cellular trajectories through a shared dynamical representation and enabling simultaneous reconstruction of regulatory interactions, attractor landscapes, and perturbational responses. A key finding is the importance of explicitly addressing the inverse problem in single-cell regulatory inference: accurately reconstructing cellular dynamics does not imply accurate recovery of causal regulatory structure. Our sensitivity analyses make this concrete. Perturbation-based driver identification was robust to the choice of prior network and regularization strength, whereas a static network score was unstable and biologically uninformative, and removing the transcription-factor scaffold entirely degraded recovery of canonical regulators. Incorporating prior biological knowledge through a scaffold substantially enhances identifiability while retaining the flexibility to discover novel interactions. An additional advantage of the scHopfield formulation is that inferred networks naturally define an energy landscape and its stability structure, enabling analysis of attractor geometry, local stability, and oscillatory behavior. In pancreatic endocrinogenesis, progenitor populations occupied more heterogeneous and dynamically flexible regions of the landscape than terminally differentiated populations, providing a quantitative counterpart to long-standing views of differentiation as movement across a structured landscape. Our perturbational analyses further suggest that perturbation biology can be viewed as a dynamical extrapolation problem: by embedding perturbations within an inferred dynamical system, scHopfield recovered known lineage regulators and generated higher-order predictions that were not apparent from static analyses. Several limitations should be considered. First, the accuracy of inferred interactions depends on the quality of RNA-velocity estimates and the completeness of biological priors. Second, the inferred networks capture effective dynamical interactions and may not map directly to physical molecular interactions. Third, recovery is faithful for systems in the model's function class; biophysical circuits that are not natively of Hopfield form are not recovered as well, a limitation we report explicitly. Fourth, predicted novel interactions require experimental validation. Future work could extend the framework to integrate multi-omic, chromatin-accessibility, spatial, and lineage-tracing data. Taken together, our results establish a unified framework in which regulatory networks, developmental trajectories, cellular attractors, and perturbational responses emerge from a common dynamical representation.

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